

Aptimizer: Master Product Roadmap & Technical Remediation Plan

From Conceptual Planning to Legally Defensible Civil Engineering AI OS | September 2026

Executive Context & Platform Audit

Aptimizer has established an impressive, unprecedented algorithmic core. Across Versions 1 to 3, the platform has built 324 catalogued capabilities spanning polygon setback derivations, multi-tower greedy and genetic layout packing, IS/NBC structural heuristics, CPM scheduling with congestion efficiency modeling, and an in-app AI assistant (APT) backed by a 74-clause Indian Standards RAG registry. However, to transition from an exploratory feasibility calculator into an indispensable, enterprise-grade software adopted by Tier-1 developers (DLF, Godrej, Prestige) and EPC giants (L&T, Shapoorji), Aptimizer must bridge critical industry gaps. This document details the **exact fixes for existing features** and the **multi-phase roadmap of what to build next**, articulated through the engineering lens of *What, How, and Why*.

Current Platform Status: 420 Features Catalogued | 324 Built & Running | 28 Partially Built | 68 Planned
Primary Gaps: CAD/BIM Interoperability (0% Built in V5), Structural Biaxial/Drift Realism, Statutory RERA Area Splits, Local Municipal DCR Presets, and 3D Earthwork Cut-and-Fill Site Grading.

Part 1: Existing Features to Fix & Exact Remediation

Before expanding into autonomous systems, the following six core modules require immediate engineering elevation to ensure absolute technical and legal defensibility.

1. Structural Column Biaxial Moments & High-Rise Foundation Selection

WHAT is broken: In backend/engineering.py (Module 1 & 3), column sizing relies solely on axial load capacity (IS 456 Cl. 39.3: $P_u = 0.4 f_{ck} A_c + 0.67 f_y A_{sc}$). In real multi-storey buildings (>15m), lateral wind and seismic shears create severe overturning moments, inducing biaxial bending (M_{ux} , M_{uy}). Sizing on axial load alone severely undersizes perimeter/corner columns. Furthermore, Module 3 prescribes isolated pad footings regardless of height, which would suffer catastrophic differential settlement on tall towers.

HOW to fix it:

- Biaxial Interaction:** Implement the Bresler interaction formulation from IS 456 Cl. 39.6:
$$\left(\frac{M_{ux}}{M_{ux1}}\right)^{\alpha_n} + \left(\frac{M_{uy}}{M_{uy1}}\right)^{\alpha_n} \leq 1.0$$

Approximate lateral moments from wind/seismic base shear distributed across the frame.
- Foundation Advisor:** Compute average contact pressure $q_{avg} = \Sigma P_{service} / \text{Footprint}$. If $q_{avg} > SBC$, automatically reject isolated footings. Prescribe **RCC Raft/Mat Foundation (IS 2950)** for moderate loads, or **Bored Cast-in-Situ Piling (IS 2911)** when building height > 45m or on soft soil, outputting required pile count and diameter.

WHY: Any chartered structural engineer will reject plans sized only for axial loads. Making sizing biaxially defensible establishes immediate professional credibility.

2. Procedural Floor Plan Hierarchy & Core Pinning

WHAT is broken: In backend/floorplan/, room layouts are generated by slicing the perimeter footprint inward. In tall residential towers, physical architecture is strictly dictated by the **central vertical structural core** (shear walls, lifts, fire escape stairs, and MEP risers). Slicing rooms without a locked core produces unbuildable apartment layouts.

HOW to fix it: Invert the generator hierarchy. Pin the core first: (1) Two fire-rated staircases ($\geq 1.5\text{m}$ tread per NBC Part 4 Cl. 4.3), (2) Passenger and stretcher fire lift shafts with smoke-stop lobbies, (3) Four dedicated vertical MEP shafts (electrical, PHE wet riser, stormwater, ventilation). Only once the core is frozen do apartment units expand outward into cruciform, linear, or L-shaped wings.

WHY: Real MEP risers and structural shear cores must align vertically from foundation to terrace without offset. Locking the core guarantees structural viability.

3. Statutory Area Accounting & RERA Carpet Schedule

WHAT is broken: backend/engine.py relies on legacy 'Super Built-up Area' loading factors (25%). Under the Indian Real Estate (Regulation and Development) Act 2016 (RERA), selling apartments on super built-up area is illegal; sales must occur strictly on **RERA Carpet Area**.

HOW to fix it: Implement Section 2(k) of RERA 2016:

RERA Carpet Area = Usable Internal Floor Area + Internal Partition Wall Thicknesses

(Explicitly excluding: External walls, service shafts, exclusive balconies/verandahs, and common areas).

Provide a statutory RERA schedule table per unit type alongside sanction drawings.

WHY: Real estate developers cannot launch projects or secure project finance without RERA-certified carpet area schedules.

4. BOQ Substructure vs. Superstructure Split & CPWD DSR Mapping

WHAT is broken: backend/takeoff.py and engine.py aggregate concrete and steel into uniform bulk totals. In construction finance, substructure works (deep excavation, shoring, raft, basement waterproofing) carry completely different risk, rates, and cash flow curves than typical superstructure floors.

HOW to fix it: Split BOQ into two distinct packages: **Package A: Substructure** (earthwork, retaining walls, foundation raft/piles, dewatering) and **Package B: Superstructure** (columns, shear walls, slabs, masonry, finishes). Map line items to standard 5-digit **CPWD Delhi Schedule of Rates (DSR)** codes.

WHY: Lenders, quantity surveyors, and contractors bid substructure and superstructure separately. DSR mapping enables government and institutional adoption.

5. Construction Programme Climate Windows & Tower Crane Cycles

WHAT is broken: In backend/schedule.py, CPM calculation assumes continuous output year-round and infinite vertical material hoisting. In India, monsoon months (June–September) severely disrupt excavation, external plastering, and waterproofing. Furthermore, high-rise floor cycle speed is physically governed by the **Tower Crane hook time**, not just crew size.

HOW to fix it:

- *Monsoon Blackouts:* Introduce regional monsoon blackout calendars in coastal/monsoon zones with automatic activity shifting or 50% productivity derating.
- *Crane Hook Cycle Limit:* Constrain the floor cycle: $T_{cycle} = \max(\text{IS 456 Striking Days}, \text{Total Tonnage} / \text{Crane Hourly Hoist Rate})$.

WHY: Real contractors will immediately identify an unconstrained monsoon or crane schedule as unrealistic, undermining confidence in the programme.

6. Internal Road Geometry & Fire Tender Swept-Path Verification

WHAT is broken: backend/siteplan/reserve.py generates inward road networks, but corners are sharp 90-degree polygon vertices. Under NBC 2016 Part 4, fire tenders require a minimum **9.0m inside turning radius (12.0m outside turning radius)** clear of obstructions.

HOW to fix it: Apply a fillet curve with minimum radius $R_{in} \geq 9.0m$ at every road junction and driveway bend using Shapely geometry buffers. Flag non-compliant corners where fire engines would become trapped.

WHY: Municipal Fire NOC (No Objection Certificate) is the #1 reason preliminary site sanction drawings are rejected by authorities.

Part 2: What to Do Next — Multi-Phase Roadmap

The following four-phase execution roadmap prioritizes high-leverage commercial and technical breakthroughs, progressing systematically from immediate CAD/BIM interoperability to an autonomous civil engineering platform.

Phase	Timeline	Key Deliverables	AEC Software / Tools	Commercial Impact
Phase 1 Interoperability Bridge	Weeks 1–3	1-Click Layered AutoCAD (.dxf) - OpenBIM IFC4 3D Export - ETABS (.s2k) / STAAD (.std) Export	ezdxf ifcopenshell CSI / Bentley format	Eliminates the CAD silo; enables structural consultants & architects to adopt Aptimizer instantly.
Phase 2 Civil & Municipal Rigor	Weeks 4–7	3D Earthwork Cut-and-Fill Balancer - City DCR Engines (Mumbai, BLR, HYD) - PreDCR Sanction Layer Automation	TIN / Grid Prismoidal Local Master Plans AutoDCR / PreDCR	Saves millions in site earthworks; enables automated municipal building sanction submission.
Phase 3 Headless CAD Sheets	Weeks 8–12	- Headless FreeCAD TechDraw Worker - Automated A1/A0 Sanction Drawing Sheets - Spatial 3D Redlining & Approval Gates	Headless FreeCAD Celery / Redis Three.js Pinning	Replaces manual drafting of municipal submission sheets; enables enterprise team sign-offs.
Phase 4 Autonomous Civil OS (X)	Weeks 13–18	- Multi-Agent Engineering Swarm - Live TMT/Cement Price Tender Bundler - Drone Orthophoto vs CPM Verification	Agent Swarms Wholesale Indices Photogrammetry	Delivers the complete Aptimizer X vision: autonomous design optimization and real-time site monitoring.

Deep Dive: Phase 1 — The Interoperability Bridge (Weeks 1–3)

Deliverable 1.1: 1-Click Layered AutoCAD Export (ezdxf)

- What:** Instant download of a production-ready .dxf drawing from any project state.
- How:** Implement backend/cad_export.py using ezdxf. Create dedicated layers: A-SITE-BNDY (property polygon), A-SITE-STBK (setback limits), A-BLDG-FP (tower footprints), S-GRID (column grid lines with bubble labels A, B, C / 1, 2, 3), S-COLS (hatched column rectangles), and A-TEXT-DIMS (linear dimensions and room carpet area tags).
- Why:** 100% of Indian architecture firms use AutoCAD. Giving them clean DXF files eliminates redrawing from scratch.

Deliverable 1.2: Certified OpenBIM IFC4 Export (ifcopenshell)

- What:** Certified 3D BIM export compatible with Revit, ArchiCAD, and Navisworks.
- How:** Use pure Python ifcopenshell to build IfcProject → IfcSite → IfcBuildingStorey. Extrude slabs as IfcSlab and columns as IfcColumn, attaching custom property sets (*Pset_CivilEngineering* with loads and concrete grades).
- Why:** Unlocks Version 5 (BIM Readiness) with zero desktop CAD dependencies.

Deliverable 1.3: ETABS / STAAD.Pro Analytical Script Export

- What:** Generate .s2k (ETABS) or .std (STAAD) text scripts containing nodes, frames, and load cases.
- Why:** Structural engineers spend 2–3 days setting up an analytical model. Aptimizer will generate it in 500 milliseconds.

Deep Dive: Phase 2 — Civil & Municipal Sanction Engine (Weeks 4–7)

Deliverable 2.1: 3D Earthwork Cut-and-Fill Balancer

- What:** Automatically calculate excavation vs fill volumes and optimize finished site formation levels.
- How:** Superimpose 3D terrain grid from GIS elevation data over building pads and internal roads. Calculate cut/fill using the prismoidal method: $\Delta V = \sum (Z_{existing} - Z_{formation}) \times \Delta x \Delta y$. Compute the optimal datum level Z_0 where $V_{cut} \approx V_{fill} \times 1.15$ (soil bulking factor).
- Why:** Hauling excess soil off-site or importing fill costs lakhs of rupees. Zero-net earthwork balance is a massive cost-saver.

Deliverable 2.2: City DCR Presets & PreDCR Sanction Export

- What:** Pre-configured rules for Mumbai (DCPR 2034 Fungible FSI), Bengaluru (BBMP/BDA Master Plan), and Hyderabad (GO 168).
- How:** Provide an export mode formatting DXF layers with designated PreDCR polylines (_BLDG_BOUNDARY, _NET_PLOT, _PROP_WORK).
- Why:** Over 80% of municipal corporations in India use AutoDCR/PreDCR for automated building sanction.

Deep Dive: Phase 3 — Headless FreeCAD & Drawing Sheet Automation (Weeks 8–12)

Deliverable 3.1: Headless FreeCAD TechDraw Sanction Sheet Worker

- *What:* Background generation of multi-page, formatted A1/A0 architectural drawing sheets.
- *How:* Run headless FreeCAD inside an asynchronous Celery worker container. Using FreeCAD's TechDraw workbench, import the building geometry, slice horizontal floor plans at $Z = 1.2\text{m}$, generate vertical cross-sections, attach title blocks with developer name, project coordinates, and area tables, and render publication-grade PDF sheets.
- *Why:* Offloads heavy drawing projection from the user's browser, automating days of manual drafting work.

Deliverable 3.2: Multi-Party 3D Spatial Redlining & Approval Gates

- *What:* Enterprise collaboration workflow allowing stakeholders to pin comments directly onto 3D building elements and enforce statutory sign-offs.
- *Why:* Fulfills the Version 4 Enterprise Collaboration promise.

Deep Dive: Phase 4 — Optimizer X: Autonomous Civil Engineering OS (Weeks 13–18)

Deliverable 4.1: Multi-Agent Engineering Swarm

- *What:* Replace the single-turn APT chat with an autonomous 4-agent background swarm: (1) *Structural Sentinel* (flags drift/slenderness), (2) *Sanction Watchdog* (monitors fire radii & setbacks), (3) *Commercial Guardian* (tracks cost/IRR deltas), and (4) *Carbon Officer*.

Deliverable 4.2: Dynamic Material Procurement & Tender Generator

- *What:* Connect to real-time wholesale commodity price indices (TMT steel, cement) and auto-generate FIDIC / Indian Standard tender packages.

Deliverable 4.3: Drone Survey Photogrammetry vs. CPM As-Built Verification

- *What:* Ingest site drone point clouds (.las) and compare completed slab heights against backend/schedule.py milestones.

Part 3: Technology Strategy & The FreeCAD Question

A common technical proposal is: *'Can we fix all 3D features, floor plans, and CAD tools by embedding FreeCAD?'*
Here is the definitive architectural analysis of why a **hybrid decoupled architecture** is mandatory.

Component Layer	Recommended Technology	Why NOT Pure FreeCAD?	Core Benefit to Aptimizer
Interactive Client UI (3D Orbit, Floor Edit)	React 19 + Three.js HTML5 Canvas / SVG	FreeCAD is a C++/Qt desktop app. It cannot run responsively in a browser at 60fps.	Instant in-browser rendering; zero install; smooth, modern web user experience.
Instant 2D CAD Export (Drawings, Grids)	Pure Python ezdxf (In FastAPI Gateway)	FreeCAD takes 2–3 seconds just to boot Python modules; ezdxf exports in < 40ms.	Blazing fast 1-click DXF export with zero container overhead.
Certified OpenBIM (3D IFC Models)	Pure Python ifcopenshell (In FastAPI Gateway)	FreeCAD's Arch workbench uses ifcopenshell under the hood anyway.	Generates IFC4 building models directly in Python with full property sets.
Complex Drawings & Boolean Solids	Headless FreeCAD Worker (Celery / Redis / Docker)	FreeCAD is ideal here! It has Open CASCADE solid booleans and TechDraw projection.	Leverages FreeCAD's true strength (A1/A0 sheets, terrain booleans) without slowing down the web app.

- Strategic Summary:
1. **Immediately fix the 6 existing modules:** Column biaxial moments, raft/pile foundations, core-first floor plates, RERA carpet areas, substructure BOQ splits, and monsoon/crane scheduling limits.

2. **Execute Phase 1 immediately:** Ship ezdxf 2D CAD and ifcopenshell IFC4 BIM export. This turns Aptimizer into an interoperable commercial platform overnight.

3. **Deploy Headless FreeCAD as an asynchronous worker:** Use it strictly for heavy A1/A0 sanction drawing sheets and 3D terrain Boolean operations, keeping your frontend lightning-fast.